

Active Management of IO Beta

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IO Beta Allocation Strategy: December 2015

Summary

- ▶ **Actual Allocation:** 22%
- ▶ **Model 1 (IO Px/Cpn not duration-adjusted):** 20%.
(Last Month: 5%)
- ▶ **Model 2 (IO Px/Cpn duration-adjusted):** 6%. (Last Month: -1%)

Active Management of IO Beta

Investigate if we can design an **active** strategy for investing in IO Beta that outperforms a **passive** strategy that is always 100% invested.

Metrics and Set-up

- ▶ Create a **Hedged IO** index
 - ▶ Hedge various IOS with current coupon TBA using model durations (Barclays)¹
 - ▶ Index is created by using monthly market value weights for each IOS based on the market weight of the corresponding cohort
 - ▶ Ginnie Mae IOS are excluded
- ▶ Simulating Passive IO Beta: Total Return and Sharpe Ratio of Hedged IO index from March 2011 - November 2015
- ▶ Simulating Active IO Beta: Dynamic allocation to Hedged IO index based on relative value metrics

¹Note that we do *not* hedge out the curve risk and volatility risk associated with an individual IOS.

Performance of Passive IO Beta Strategy: Mar 2011 - Nov 2015

- ▶ **CAGR:** 9.8%
- ▶ **Ann. Volatility:** 16.7%
- ▶ **Sharpe Ratio:** 0.6
- ▶ **% Positive Months:** 60
- ▶ **Best Month:** 11.6%
- ▶ **Worst Month:** -12.9%
- ▶ **Max Drawdown:** -23.5%
- ▶ **Drawdown Period:** June 2011 - March 2013

Framework for Active IO Beta Strategy

- ▶ Regress Hedged IO Index returns against RV metrics (1m lag):
 - ▶ IO Index Price/Coupon²
 - ▶ Projected 1-month Carry³
 - ▶ Convexity: 60-day moving average of realized volatility on the current coupon
 - ▶ Prepayment Risk: Incentive for the Mortgage Universe
 - ▶ Yield Curve Slope: 10s-2s slope
- ▶ Calculate z-score of projected return and allocate to Beta (Net IO) based on score

²While OAS would also be useful, in practice dealer model updates are not always appropriately back-filled which makes it challenging to get a consistent time series of values.

³This is not available on a historical basis so we use realized carry as a proxy.

Active IO Beta Strategy: Regression Results

- ▶ Significant Variables
 - ▶ High: IO Index Price/Coupon, Prepayment Risk
 - ▶ Medium: Realized Volatility
 - ▶ Low: Realized Carry, Yield Curve Slope, Index OAS
- ▶ Final Model
 - ▶ Proj. Return $\sim P_x/C_{pn} + C_{nvx} + \text{PrepayRisk} + \text{Carry}$

Active IO Beta Strategy: Beta Allocation

- ▶ Solve for $k = a + bz$ ($z = z\text{-score}$) such that we optimize the Sharpe Ratio of the strategy $k \times \text{Hedged IO Return}$.
- ▶ Set $\text{IO Beta}_{\max} = 100\%$ and $\text{IO Beta}_{\min} = -0.25\%$
- ▶ Rescale k so that it is (more or less) mapped to these boundaries.

Active IO Beta Strategy: Two Beta Allocation Models

- ▶ Model 1: Don't adjust IO Index P_X/C_{pn} for Interest Rates. $R^2 = 0.29$.
- ▶ Model 2: Adjust IO Index P_X/C_{pn} for Interest Rates.⁴ $R^2 = 0.20$.

⁴We estimate that the P_X/C_{pn} ratio moves by about 0.14 for each 10bps move in the current coupon.

Active IO Beta Strategy: Model 1 Return Profile

- ▶ **CAGR:** 10.0%
- ▶ **Ann. Volatility:** 6.0%
- ▶ **Sharpe Ratio:** 1.6
- ▶ **% Positive Months:** 73
- ▶ **Best Month:** 9.7
- ▶ **Worst Month:** -1.0%
- ▶ **Max Drawdown:** -1.6%
- ▶ **Drawdown Period:** November 2011 - March 2012

Active IO Beta Strategy: Model 2 Return Profile

- ▶ **CAGR:** 8.0%
- ▶ **Ann. Volatility:** 6.3%
- ▶ **Sharpe Ratio:** 1.3
- ▶ **% Positive Months:** 71
- ▶ **Best Month:** 9.5
- ▶ **Worst Month:** -2.9%
- ▶ **Max Drawdown:** -3.6%
- ▶ **Drawdown Period:** November 2011 - March 2012

Track Record of IO Beta Allocation Strategies

	IO Beta Allocation			Monthly Hedged Return				Yearly Return			
	Fund	Model1	Model2	100% Beta	Fund	Model1	Model2	100% Beta	Fund	Model1	Model2
Jan-15	26%	-8%	-10%	-11.8%	-3.1%	0.9%	1.2%	9.2%	1.8%	5.1%	5.1%
Feb-15	25%	43%	66%	11.2%	2.8%	4.8%	7.4%				
Mar-15	21%	-3%	-44%	-0.7%	-0.1%	0.0%	0.3%				
Apr-15	15%	-18%	-12%	4.9%	0.7%	-0.9%	-0.6%				
May-15	22%	-22%	-25%	1.2%	0.3%	-0.3%	-0.3%				
Jun-15	31%	-10%	-13%	4.6%	1.4%	-0.5%	-0.6%				
Jul-15	33%	-20%	-41%	-1.6%	-0.5%	0.3%	0.7%				
Aug-15	28%	-30%	-13%	0.1%	0.0%	0.0%	0.0%				
Sep-15	24%	-41%	-25%	-1.2%	-0.3%	0.5%	0.3%				
Oct-15	26%	-2%	12%	0.2%	0.0%	0.0%	0.0%				
Nov-15	21%	5%	0%	2.4%	0.5%	0.1%	0.0%				
Dec-15	22%	20%	6%								

Evolution of IO Beta Allocation RV Variables

Date	MthlyHedgedReturn	OAS	Carry	RealizedVol	YCSlope	PrepayRiskPrem	PxCpnMult	UB1m	RefiPctPool	RefiPctLoan	RefIndex	CC	CCChanges
1/30/2015	-11.821	65	1.1038	60.22	120.8	102.97	4.78	0.1713	0.6626	0.6558	2585	2.45	-38.12
2/27/2015	11.1577	-87	1.6224	73.25	137.6	88.62	5.51	0.173	0.611	0.5614	1740	2.74	28.39
3/31/2015	-0.7	-97	0.9897	78.46	137.9	92.38	5.37	0.1763	0.642	0.6199	1942	2.66	-7.56
4/30/2015	5.207	28	0.2384	69.09	146.7	80.98	5.56	0.181	0.6364	0.6089	1559	2.75	8.7
5/29/2015	1.2501	27	0.3672	68.4	149.2	72.75	5.66	0.184	0.5495	0.4952	1234	2.81	6.45
6/30/2015	4.679	10	0.3215	75.45	169.8	50.56	6.16	0.1865	0.45	0.397	1246	3.07	26.07
7/31/2015	-1.5451	14	0.1851	82.06	153.1	66.46	5.93	0.1918	0.4991	0.4227	1418	2.92	-15.78
8/31/2015	0.1421	21	0.4859	76.13	146.2	67.72	5.92	0.1986	0.5478	0.5324	1546	2.93	1.89
9/30/2015	-1.2489	40	0.6095	67.08	141.5	70.97	5.7	0.193	0.5189	0.4886	1590	2.8	-13.25
10/30/2015	0.1931	42	0.5574	66.44	141.6	68.26	5.69	0.192	0.5789	0.5526	1535	2.84	4.25
11/30/2015	2.3941	3	0.5458	57.52	128.7	57.14	5.89	0.243	0.4759	0.4481	1489	2.95	10.3